MILITARY SPECIFICATION

Scale, Beam Indicating: Metric System; Counter-Mounted

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a triple-beam, low-form, metric system scale having a basic 610-gram beam capacity augmented by prescribed sets of attachment counterpoise weights to establish the size classifications specified herein.

1.2 Classification. The scales shall be of the following sizes and styles, as specified (see 6.2):

Size 1 - 1,610-gram capacity.
Size 2 - 2,110-gram capacity.
Size 3 - 2,610-gram capacity.

Style A - With flat plate-type weighing platform.
Style B - With pan-type load receiving element.
Style C - With scoop-type load receiving element.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified (see 6.2), the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

PPP-B-1122 - Balances, Scale, and Accessories, Packaging and Packing of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6670

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-147 - Palletized Unit Loads.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as Department of Defense (DoD) adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet and Strip.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL BUREAU OF STANDARDS (NBS)

H44 - Specifications, Tolerances, and Regulations for Commercial Weighing and Measuring Devices.

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The beam scales (see 6.6.1), referred to as scales, shall be counter-mounted, triple-beam, low-form balances consisting essentially of a base, weighbeam assembly, and in accordance with the style specified, a load-receiving plate, pan, or scoop. Attachment counterpoise weights specified herein shall be furnished with each scale.

3.2 First article. When specified (see 6.2), the contractor shall furnish a scale for first article inspection and approval (see 4.2.1 and 6.4).

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3.3 Technical definitions. Technical definitions used throughout this specification are defined in 6.6.

3.4 Standard commercial product. The scales shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the scale being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.6.1 Stainless steel. Stainless steel specified herein for load receivers shall conform to ASTM A167 type 301.

3.7 Design. The scales shall be designed to meet the applicable performance requirements and basic acceptance tolerances of NBS H44 as converted in table I of this specification and to the metric system. The scales shall be designed to support, without permanent deformation or misalinement, a 2,625-gram load or a load equal to 125 percent of the nominal capacity (see 6.6.4), whichever is greater. The scales shall be capable of accurately registering weight indications irrespective of repeated manipulation of any scale element, as in normal usage, including: (1) repeated application and removal of maximum loads and attachment weights, (2) repeated operation of locking devices, or (3) repeated displacement of all weighbeam (see 6.6.2), poises to the limit of each beam scale. The sensitivity (see 6.6.6), of the scales under all conditions of use specified herein shall be 0.10 gram or less and the sensibility reciprocal (see 6.6.5) shall not exceed 3.5 grams. Design and performance requirements shall be tested as specified in section 4.

3.8 Details of components.

3.8.1 Base. The base shall be a sturdy metal casting with a moisture- and acid-resistant finish. The base shall be equipped at the beam end with a post

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TABLE I. Basic acceptance tolerances for under-registration and overregistration

Test loads, grams				Basic acceptance tolerances
Fraction of full load	Size 1	Size 2	Size 3	(grams)
1/8	200	263	325	0.9
1/4	400	525	650	0.9
3/8	600			0.9
3/8		789	975	1.8
1/2	800	1,050	1,300	1.8
3/4	1,200	1,575		1.8
3/4			1,950	2.7
full load	1,600	2,100	2,600	2.7

graduated, as required, to provide a definite and clear zero position and to indicate an out-of-balance condition. The post shall be equipped with stops so located, as to permit a total weighbeam travel between stops of not less than 0.4 inch. The base shall be formed with smooth rounded surfaces to prevent undue accumulation of dust or spilled material.

3.8.2 Weighbeam. The weighbeam on all size scales shall consist of three bars: (1) a 500-gram capacity beam notched at 100-gram intervals, (2) a 100-gram capacity beam notched at 10-gram intervals, and (3) a 10-gram capacity, front, smooth, fractional bar (see 6.6.3), graduated in increments of 0.10 gram. The individual beam shall be tiered from front to rear to increase scale readability and poise visibility. The weighbeam shall be equipped with a threaded zero adjusting nut at the loading end. Means shall be provided to dampen beam oscillations after application or removal of a load.

* 3.8.3 Pivots and bearings. Bearings shall be self alining and shall be hardened steel, ceramic, or polished agate at the option of the contractor. Pivot knife edges shall be sharp and straight and shall be hardened steel. Each scale shall be equipped with four bearings and two pairs of pivots. Bearings shall be protected by dust covers or equivalent enclosures to minimize contamination and corrosion of the precision bearing surface.

3.8.4 Load receiving elements.

3.8.4.1 Style A. Plates for style A scales shall be stainless steel and shall be 6 inches ±1/2-inch in diameter. Plates shall be flat except that a raised peripheral lip not exceeding 1/4-inch in height will be acceptable.

3.8.4.2 Style B. Pans for style B scales shall be stainless steel and shall be 6 inches ±1/2-inch in diameter by 1 inch ±1/4-inch in depth.

3.8.4.3 Style C. Scoops for style C scales shall be not less than 10 inches long by 6 inches wide with a reinforced edge.

3.8.4.4 Cover. A translucent or opaque plastic fitted cover shall be furnished with each scale.

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* 3.8.5 Attachment weight set. Each scale shall be furnished with a set of attachment counterpoise weights. The sets shall consist of the following weights:

a. Size 1: Two 500-gram weights.
b. Size 2: One 1,000 gram and one 500-gram weight.
c. Size 3: Two 1,000 gram and one 500-gram weights.

Each weight shall be legibly marked to indicate the equivalent load capacity. Means shall be provided on the base for storage of the weights when not in use.

3.9 Accessories. When specified (see 6.2), either a tare weight assembly or a specific gravity assembly or both shall be furnished with each scale.

3.9.1 Tare weight assembly. The tare weight assembly shall consist of a separate tare beam and poise, or a tare counterpoise weight. The tare assembly shall be adjustable within a range up to at least 200 grams.

3.9.2 Specific gravity assembly. The specific gravity assembly shall consist essentially of a stand or counter-clamp and rod for supporting the scale above the counter surface at a height of at least 10 inches.

3.10 Marking. The actual capacity shall be clearly and permanently marked on each scale.

3.11 Finish. The scales shall have all surfaces finished in accordance with the manufacturer's standard practice.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one complete scale when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.4 and the tests of 4.5. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

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* 4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

* 4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. All scales offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.3.1 Sampling for examination. Examination shall be based on inspection level II and an Acceptable Quality Level (AQL) of 2.5 percent defective.

4.3.2 Sampling for tests. Tests shall be based on inspection level S-3 and an AQL of 4.0 percent defective.

* 4.4 Examination. Each scale selected shall be examined for compliance with the requirements specified in Section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement of presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Tests.

4.5.1 Increasing-load test. Apply and remove weights of 1/8, 1/4, 3/8, 1/2, 3/4, and full load in ascending order (see Table I). Record the readings obtained with scale indicating elements and with counterpoise weights, when used. Compare readings with test weights using basic acceptance tolerances as required to verify acceptability of the scale.

4.5.2 Sensibility reciprocal (SR) test. Determinations of SR shall be made in conjunction with the increasing load test at zero load, half load, and full load to verify compliance with 3.7.

4.5.3 Shift test. Apply one-half of full load as specified in Table I successively at not less than four equidistant points between the center and edge of the load-receiving element. On style C scales use a pan or plate in place of the scoop. Failure to comply with basic acceptance tolerances shall be cause for rejection.

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4.5.4 Repeatability test. Apply and remove full load weight five times, returning poises to zero position after each weighing. Note position of weighbeam with respect to balance position each time. Failure of the scale to meet the applicable basic full load acceptance tolerance of Table I shall be cause for rejection.

4.5.5 Overload test. Load the scale with a weight equal to 2,625 grams for sizes 1 and 2, and 3,250 grams for size 3 with the scale counter-balanced to full load. Remove all weights, return poises to zero position, and apply the full load test weight specified in Table I. Observe scale reading at balance position. Repeat five times. Failure of the scale to register the test weight within applicable tolerances of Table I for full loads or any observable deformation or misalinement of components shall be cause for rejection of the scale. The overload test may be conducted in conjunction with the repeatability test at the option of the contractor.

* 4.6 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of PPP-B-1122.

5. PACKAGING

* 5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of PPP-B-1122 with the level of preservation and the level of packing as specified (see 6.2). When specified (see 6.2), palletization in accordance with MIL-STD-147 shall be required.

6. NOTES

* 6.1 Intended use. The scales covered by this specification are intended for general purpose laboratory work and as basic equipment for soil, aggregate, asphalt, and concrete testing laboratories.

* 6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Size and style of scale required (see 1.2).
c. Specification and standards (see 2.1.1).
d. When a sample scale is required for first article inspection and approval (see 3.2, 4.2.1, and 6.4).
e. When a tare weight assembly, a specific gravity assembly, or both are required (see 3.9).
f. Level of preservation and level of packing required (see 5.1).
g. When palletization is required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 52.227-7031 of the Federal Acquisition Regulations (FAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the

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provisions of FAR 52.227-7031 are not invoked, the data shall be delivered in accordance with the contract requirements.

* 6.4 First article. When a first article inspection is required (see 3.2 and 6.2), the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one complete scale unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.6 Definitions. The following definitions shall be mandatory for use by the contractor and the aquisitioning agency in the interpretation of the requirements of this specification.

6.6.1 Beam scale. A scale on which the weights of loads are indicated solely by means of one or more weighbeam bars either alone or in combination with counterpoise weights.

6.6.2 Weighbeam. An element comprising one or more bars (three on a triple-beam scale) equipped with movable poises (weights) and, under this specification, means for applying counterpoises (hanging weights at the tip of the weighbeam).

6.6.3 Fractional bar. A weighbeam bar of relatively small capacity for obtaining weight indications intermediate to the notches of the next highest main bar.

6.6.4 Nominal capacity. The largest weight indication obtainable by using all reading elements in combination with removable attachment weights furnished with the scale is its normal capacity. Normally, the capacity of fractional bars is included in the nominal capacity only when the fractional bar exceeds 2-1/2 percent of the sum of the capacities of the remaining reading elements. Under this specification, however, the fractional bar is included in the nominal capacity to be consistent with Military supply practices and commercial size designations.

6.6.5 Sensibility reciprocal. The sensibility reciprocal of scales covered by this specification, is the minimum change in load required to alter the position of rest of the weighbeam from midway between limiting stops to either limit of motion.

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6.6.6 Sensitivity. The sensitivity of a triple-beam scale is an indication of the smallest weight or weight differential which can be detected and quantified, using the fractional poise and thus is normally equivalent to the smallest graduation on the fractional bar.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-19423D	SCALE BEAM INDICATING:METRIC SYSTEM; COUNTER-MOUNTED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
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a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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